MACHINE LEARNING COURSEWORK

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1. Unsupervised Learning

1.1 Problem Statement

Living costs are influenced by a multitude of factors such as the cost of enjoying a meal, the cost of taking a taxi, the cost of rent, and more. As such, different cities have different costs of living. The average price levels across OECD countries were 35.6% higher in January 2026 than before COVID-19 (OECD, 2026). Being able to predict and understand global living costs would assist numerous stakeholders, like households and policy makers, in making informed decisions regarding finances and relocation. Therefore, we aim to implement unsupervised learning techniques to group the cities into clusters and interpret cities with similar costs of living.

1.2 Research Questions

RQ1) What is the optimal number of clusters to group cities?
RQ2) What are the key factors differentiating the cost-of-living clusters across cities?
RQ3) What are the principal dimensions of variation in global living costs?

1.3 Dataset and Exploratory Data Analysis (EDA)

This dataset is known as “Global Cost of Living” which consists of 58 columns and information from approximately 5000 cities globally. Columns x1 to x55 are price variables in USD. There are some missing values and no duplicate values in the dataset.

Variable	**Description**
city	Name of the city
country	Name of the country
x1	Meal, Inexpensive restaurant
x2	Meal for 2 people, Mid-range restaurant, Three course
x3	McMeal at McDonalds (or equivalent combo meal)
x4	Domestic Beer (0.5 litre draught, in restaurants)
x5	Imported Beer (0.33 litre bottle, in restaurants)
x6	Cappuccino (regular, in restaurants)
x7	Coke/Pepsi (0.33 litre bottle, in restaurants)
x8	Water (0.33 litre bottle, in restaurants)
x9	Milk (regular, 1 litre)
x10	Loaf of Fresh White Bread (500g)
x11	Rice (white, 1kg)
x12	Eggs (regular, 12)
x13	Local Cheese (1kg)
x14	Chicken Fillets (1kg)
x15	Beef Round (1kg or equivalent Back Leg Red Meat)
x16	Apples (1kg)
x17	Banana (1kg)
x18	Oranges (1kg)

Variable	Description
x30	Taxi Start (normal tariff)
x31	Taxi 1km (normal tariff)
x32	Taxi 1hour waiting (normal tariff)
x33	Gasoline (1 liter)
x34	Volkswagen Golf 1.4 90 KW Trendline (Or Equivalent New Car)
x35	Toyota Corolla Sedan 1.6l 97kW Comfort (Or Equivalent New Car)
x36	Basic (Electricity, Heating, Cooling, Water, Garbage) for 85m2 Apartment
x37	1 min. of Prepaid Mobile Tariff Local (No Discounts or Plans)
x38	Internet (60 Mbps or More, Unlimited Data, Cable/ADSL)
x39	Fitness Club, Monthly Fee for 1 Adult
x40	Tennis Court Rent (1 Hour on Weekend)
x41	Cinema, International Release, 1 Seat
x42	Preschool (or Kindergarten), Full Day, Private, Monthly for 1 Child
x43	International Primary School, Yearly for 1 Child
x44	1 Pair of Jeans (Levis 501 Or Similar)
x45	1 Summer Dress in a Chain Store (Zara, H&M, …)
x46	1 Pair of Nike Running Shoes (Mid-Range
x47	1 Pair of Men Leather Business Shoes
x48	Apartment (1 bedroom) in City Centre

x19	Tomato (1kg)
x20	Potato (1kg)
x21	Onion (1kg)
x22	Lettuce (1 head)
x23	Water (1.5 liter bottle, at the market)
x24	Bottle of Wine (mid-range, at the market)
x25	Domestic Beer (0.5 liter bottle, at the market)
x26	Imported Beer (0.33 liter bottle, at the market)
x27	Cigarettes 20 pack (Marlboro)
x28	One-way Ticket (local transport)
x29	Monthly Pass (regular price)

x49	Apartment (1 bedroom) Outside of Centre
x50	Apartment (3 bedrooms) in City Centre
x51	Apartment (3 bedrooms) Outside of Centre
x52	Price per Square Meter to Buy Apartment in City Centre
x53	Price per Square Meter to Buy Apartment Outside of Centre
x54	Average Monthly Net Salary (After Tax)
x55	Mortgage Interest Rate in Percentages (%), Yearly, for 20 Years Fixed-Rate
data_quality	0 if Numbeo considers that more contributors are needed to increase data quality, else 1

Table 1: Unsupervised learning dataset and variable description

Due to missing values, the dataset had to be preprocessed. For variables with more than 30% missing values, the variables were dropped from the dataset. The variables dropped are x28, x29, x40, x43, x51, x52, and x53. After that, the remaining missing values in the remaining columns were replaced with its median.

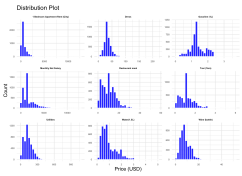


Figure 1. Distribution plots of representative variables

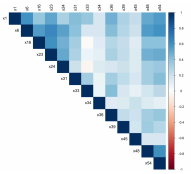


Figure 2. Correlation matrix (heatmap)

Before diving into unsupervised learning techniques, exploratory data analysis (EDA) was conducted. As shown in Figure 1, all nine distributions are skewed to the right. This means that most cities cluster at lower price levels with a small number of cities pulling the distribution towards the upper tail. The rent for a 1 bedroom apartment in the city centre has the most right skewed distribution with rent as affordable as USD21.70 in Puttalam, Sri Lanka to the most expensive rent at USD12608.83 in Kermanshah, Iran. Gasoline has a relatively normal distribution compared to the other variables. This could be due to regulation of fuel prices causing less dispersion of its prices.

The correlation heatmap shows the relationships between variables. As seen in Figure 2, most of the heatmap is in various shades of blue. Most of the variables are positively correlated. The stronger positive correlated relationships can be seen between x1 and x6, x6 and x23, x48 and x54 to name a few. x1(restaurant meal) and x6(cappuccino in restaurant) having strong positive correlations suggests that people that spend more eating a meal at restaurants tend to order more expensive cappuccino at restaurants. This is generally expected as both variables are related to spending at restaurants. Some variables have little to no correlation as seen by the white squares in Figure 2, such as x16 and x33, x24 and x33. x16 (apple) and x33 (gasoline) have no relationship, suggesting that the prices of these 2 variables behave independently of one another. This is expected as x16 and x33 belong to different categories, food and energy respectively, and hence, their prices are influenced by different factors. Therefore, with multicollinearity due to the presence of many positively correlated variables, we should do principal component analysis (PCA).

1.4 Analysis

1.4.1 Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique that is used to reduce the number of variables in the dataset while retaining as much information as possible. PCA transforms the originally correlated variables to be uncorrelated variables known as principal components (PCs).

Prior to performing PCA, the dataset had to be transformed. A new dataframe was created to include only price variables so the 3 variables, city, country, and data_quality, were dropped. Next, the data was standardised to ensure each variable contributes equally to the analysis. PCA was implemented with this transformed dataset.

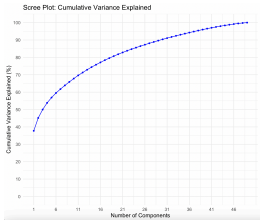


Figure 3. PCA scree plot

This scree plot shows the cumulative variance explained by the PCs. From Figure 3, PC1 explains approximately 37% of the cumulative variance on its own and the first 19 PCs explain more than 80% of the cumulative variance. This result from PCA allows us to better understand how different variables contribute to the cumulative variance.

1.4.2 K-Means Clustering

K-means clustering is an unsupervised learning technique used to group data into clusters based on its similarity. It aims to minimise the within cluster variance.

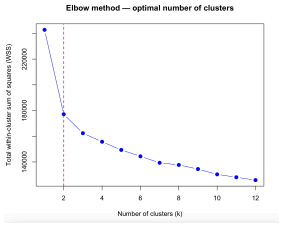


Figure 4. Elbow method

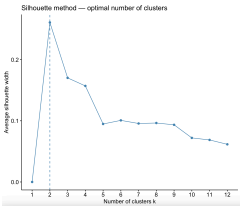


Figure 5. Silhouette method

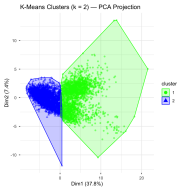


Figure 6. Cluster plot

Using statistical techniques such as the elbow and silhouette method, the optimal number of clusters can be determined. According to Figure 4 and 5, the optimal number of clusters is two. Using the optimal number of clusters, it projected onto the first 2 PCs as seen in Figure 6. Along Dim1, the two clusters are separated into positive and negative values. The green circles cluster occupies the right side of the plot with positive values in Dim1 while the blue triangles cluster occupies the left side of the plot with zero and negative values in Dim1. This suggests that the two clusters differ in their price levels, separated to expensive and affordable cities to live in.

1.4.3 Hierarchical Clustering

Hierarchical clustering is an unsupervised learning technique that groups data into clusters through building a hierarchical tree-like structure. Instead of choosing a fixed number of clusters at the start, it shows step-by-step the way clusters are formed. Data points continuously merge with its closest clusters until it becomes one cluster at the top. The result would be shown as a dendrogram as seen below in Figure 7. The Euclidean distance and the height of the branches measure the similarity between cities.

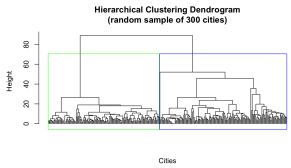


Figure 7. Dendrogram

Due to the dendrogram being unreadable when all 4700 cities' dataset were used, a random sample of 300 cities' dataset was used instead. From the dendrogram in Figure 7, the optimal number of clusters is two as shown by the green and blue structures. This is consistent with the results in K-means clustering using the elbow and silhouette methods.

1.5 Results

RQ1) What is the optimal number of clusters to group cities?
The optimal number of clusters to group cities was determined to be two using the Elbow and Silhouette methods in K-means clustering and the dendrogram in hierarchical clustering. The elbow plotted the total within-cluster sum of squares against k, resulting in a sharp decline between k=1 and k=2 before the decline turns gradual for the rest of k. The silhouette method measures

how well each city fits into its assigned cluster relative to its neighbouring clusters and the silhouette score peaked at k=2. The dendrogram shows two clusters from its blue and green structures. This suggests a clear divide into two clusters, expensive cities clustered into one and affordable cities clustered into the other.

RQ2) What are the key factors differentiating the cost-of-living clusters across cities?
Through calculating the median value of key price variables for each cluster, it reveals significant differences between price variables. Cluster 1 (higher cost cities) had a median monthly net salary of USD2284 compared to USD777 for Cluster 2 (lower cost cities), representing a 194% difference. The difference between the cost of 1 bedroom apartment rent in the city centre is substantial too, with a median of USD800 for Cluster 1 and USD388 for Cluster 2. From PCA, leisure, food, and beverage variables contribute the most to PC1.

RQ3) What are the principal dimensions of variation in global living costs?
From Figure 3 in PCA , the scree plot shows PC1 and PC2 accounts for 37% and 7% of the total variance respectively. The first 19 PCs explain more than 80% of the overall variance. This represents a significant reduction in dimensionality from the original 55 price variables. Examining the PC1 loadings, variables like leisure, food, and beverages in restaurants contributed the most. For PC2 loadings, variables like energy, clothing, food in the market contributed the most. These two PCs capture two different sets of variables. PC1 captures the dining out and having fun lifestyle while PC2 captures the more general day-to-day costs.

2. Regression

2.1 Problem Statement

Predicting hotel room prices is one of the most commercially valuable and challenging problems in hospitality revenue management. Hotel room prices are influenced by various factors, such as the type of room, the month, and the number of adults staying in a room. Thus, hotel room prices can differ from day to day. The dynamic pricing software market grew from $3.05 billion in 2024 to a projected $3.53 billion in 2025, growing at a CAGR of 15.8% (Acropolium, 2025). The challenge comes from accurately predicting hotel room prices while taking into account its various factors. Hence, we aim to identify the most important factors affecting hotel room prices and the best regression to predict hotel room prices.

2.2 Research Questions

RQ1) What are the key factors that influence average hotel room prices per night?
RQ2) Which regression model is the best at predicting average hotel room prices per night?
RQ3) How does regularisation affect model performance and feature selection in predicting average hotel room prices per night?

2.3 Dataset and Exploratory Data Analysis (EDA)

The dataset is called “Hotel Reservations Dataset” consisting of 36275 rows and 19 columns. There are no missing values in this dataset.

Variable	Description
Booking_ID	Unique identifier of each booking
no_of_adults	Number of adults
no_of_children	Number of children
no_of_weekend_nights	Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
no_of_week_nights	Number of week nights (Monday to Friday) the guest stayed or booked to stay at the hotel
type_of_meal_plan	Type of meal plan booked by the customer
required_car_parking_space	Does the customer require a car parking space? (0-No, 1-Yes)
room_type_reserved	Type of room reserved by the customer
lead_time	Number of days between the date of booking and the arrival date

Variable	Description
arrival_year	Year of arrival date
arrival_month	Month of arrival date
arrival_date	Date of the month
market_segment_type	Market segment designation
repeated_guest	Is the customer a repeat guest? (0-No, 1-Yes)
no_of_previous_cancellations	Number of previous bookings that were canceled by the customer prior to the current booking
no_of_previous_bookings_not_canceled	Number of previous bookings not canceled by the customer prior to the current booking
avg_price_per_room	Average price per day of the reservation; prices of the rooms are dynamic (in euros)
no_of_special_requests	Total number of special requests made by the customer
booking_status	Flag indicating if the booking was cancelled or not

Table 2: Regression dataset and variable description

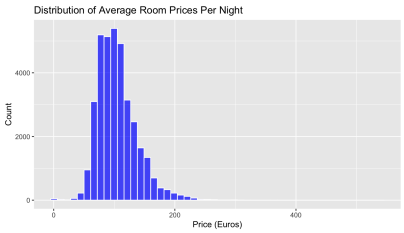


Figure 8. Distribution of Average Room Prices Per Night

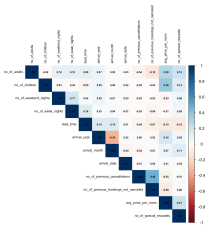


Figure 9. Correlation Matrix Heatmap

Before starting to apply regression analysis, EDA was conducted on the dataset. From Figure 8, average room prices per night is a right-skewed distribution. This means most hotel room prices are cheaper while few hotel room prices are on the higher end causing an upper tail.

Figure 9 shows that the correlations between the variables are overall weak with a few moderately strong correlations. The number of previous bookings cancelled and not cancelled have the strongest correlation in this matrix at r = 0.48 which is moderately strong correlation. This suggests that guests who have a history of previous cancellations also have a history of not canceling their hotel reservations. This reflects an overall booking activity rather than an issue. The average room prices per night have the strongest correlation with number of children at r = 0.36 followed by no_of_adults at r = 0.29. This suggests that the average hotel room price per night increases as the number of children and adults staying in the hotel room increases. This is likely due to booking for more expensive room types to accommodate more people in a hotel room.

Before starting regression analysis, the variable, booking_status, was removed from the dataset to prevent data leakage. The regression task is to predict hotel room prices at the time of booking. Whether a guest cancels or does not cancel, it is only known in the future. If booking status is included, the regression models would perform better than it would on the actual data. Hence, booking status is excluded from the dataset. A 80:20 train-test split was performed to allow the training data to build the following models. 10-fold cross validation was used on the models to obtain a more accurate prediction of hotel room price per night.

2.4 Analysis

2.4.1 Linear Regression

Linear regression is a supervised learning method used to predict a continuous numeric value through finding a best fit line in the data. It assumes a linear relationship between the dependent variables and the target independent variable through minimising the residual sum of squares.

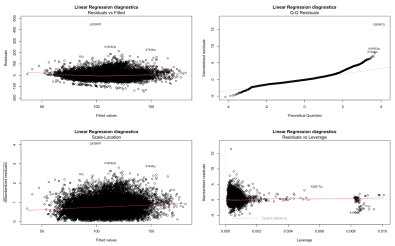


Figure 10. Diagnostic plots

Diagnostic plots were generated to check whether the linear regression model's assumptions were met. Linear regression model assumes linearity, independence, homoscedasticity, normality of residuals, and no multicollinearity.

Let's analyse Figure 10. In the "Residuals vs Fitted" plot, the residuals are broadly centred around zero across the fitted values and the red line is relatively flat. This suggests that the linearity assumption is relatively satisfied. However, there is some spreading of residuals at the upper end of the fitted values which might suggest mild heteroscedasticity where the errors for more expensive hotel rooms are larger. There are three outliers, X26087, X16452, and X7444, with large positive residuals of more than 200 EUR, suggesting underpredictions of costlier hotel rooms. In the "Q-Q Residuals" plot, the residuals generally follow the diagonal line, especially the central portion of the distribution. The tail ends deviate from the diagonal line, indicating non-normality. This is likely due to the same three outliers, X26087, X16452, and X7444, with the highest standardised residuals being more than 16. In the "Scale-Location" plot, the red line slopes upwards slightly. This suggests that there is mild heteroscedasticity which is consistent with our observation in the "Residuals vs Fitted" plot. In the "Residuals vs Leverage" plot, there were no observations outside of the Cook's distance and the majority of the observations are clustered around zero. This indicates that there is little leverage so most

observations have a similar influence on the model. Therefore, with several violations of linear regression assumptions, there is a need for testing more complex non-linear models in the subsequent analysis.

2.4.2 Ridge Regression

Ridge regression is a modified version of linear regression that adds a penalty to avoid overfitting and handle multicollinearity. L2 regularisation is the penalisation of large coefficients using lambda. The value of lambda has varying effects of the coefficients. The larger the lambda value, the closer the shrinkage is to zero. The optimal lambda value was generated through 10-fold cross validation and it minimises the cross validation RMSE. Ridge regression retains all the features in the model but with shrunken coefficients.

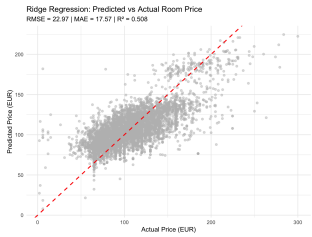


Figure 11. Ridge regression: Predicted vs Actual Hotel Room Price (EUR) plot

From Figure 11, the predicted line is red and the actual room prices are the grey points. The model had an RMSE of 22.97, MAE of 17.57, and R^2 of 0.508, which means this model explains 50.8% of the variance in hotel room prices on the test data. This indicates moderate predictive performance for this model. The grey points generally follow the red line from approximately 70 EUR to 150 EUR. Below 70 EUR, the model tends to overpredict hotel room prices. Above 150 EUR, the model tends to underpredict hotel room prices. This indicates that the model does not fully capture the factors driving certain bookings cheaper or more expensive. Hence, this model is more viable for predicting standard hotel room prices only, excluding cheaper and more expensive hotel rooms.

2.4.3 Lasso Regression

Lasso regression is a modified version of linear regression that adds a penalty to shrink some coefficients to exactly zero, removing features from the model entirely. This is called L1 regularisation. The optimal lambda value was generated with 10-fold cross-validation.

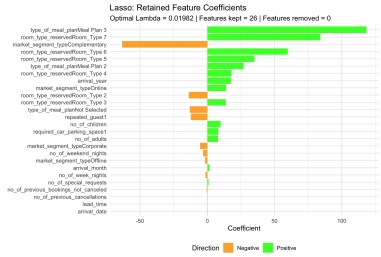


Figure 12. Retained Feature Coefficients Plot

After lasso regression with optimal lambda of 0.01982, all 26 features were retained. This suggests that there is limited redundancy among the predictors and hotel room prices are influenced by multiple combinations of variables as compared to a few dominant variables. From Figure 12, the type of meal plans and the room type reserved are the primary factors influencing hotel room prices. On the other hand, variables like, arrival date, lead time, and the number of previous cancellations, contribute little predictive power towards hotel room prices.

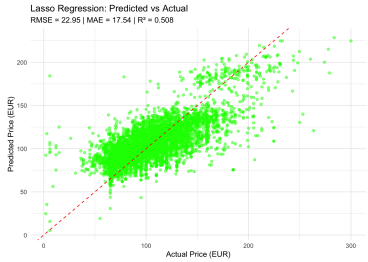


Figure 13. Lasso regression: Predicted vs Actual Hotel Room Price (EUR) plot

From Figure 13, the predicted line is red and the actual room prices are the green points. The model had an RMSE of 22.95, MAE of 17.54, and R^2 of 0.508, which means this model explains 50.8% of the variance in hotel room prices on the test data. These values are very similar to the ridge regression plot in Figure 11, suggesting and also seen in Figure 13 similar behaviour of overprediction for cheaper hotel rooms prices and underprediction for expensive hotel rooms prices. This indicates that the linear form is a constraint on the model performance and non-linear analysis must be done subsequently.

2.4.4 Random Forest

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their predictions to produce a more accurate and stable result than any single tree. Through bootstrap sampling and random feature selection, aggregate predictions are made across all trees which reduces overfitting.

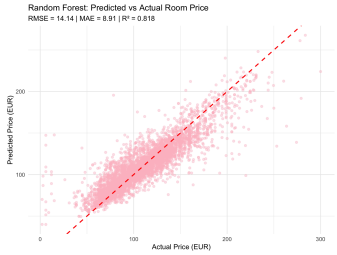


Figure 14. Random Forest: Predicted vs Actual Hotel Room Price (EUR) plot

From Figure 15, the predicted line is red and the actual room prices are the pink points. The model had an RMSE = 14.14, MAE = 8.91, and R^2 = 0.818 on the test set. The actual hotel room prices are mostly following the prediction line across the full price range. This is a much better alignment than the linear regression models. The Random Forest model is therefore identified as the best performing regression model in this study, explaining 81.8% of variance in room prices on unseen test data .

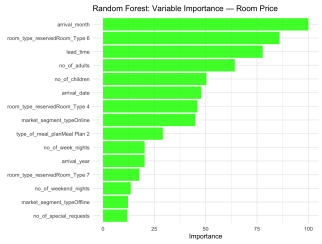


Figure 15. Feature Importance Ranking

From Figure 16, the most important predictor is the arrival month of the hotel room reservation with an importance score of 100. This suggests that hotel room prices are seasonal. This also contrasts with the findings in the lasso coefficient plot where the arrival month had a near-zero coefficient. This shows that random forest was able to capture the non-linear seasonal pattern which lasso could not capture. Room type is consistently important in both random forest and lasso, indicating that the physical characteristics of the reserved hotel room is an important predictor of hotel room rates.

2.5 Results

RQ1) What are the key factors that influence average hotel room prices per night?
Through lasso coefficient analysis and random forest feature importance, the room type, arrival month, and number of guests, are the key factors influencing nightly hotel room rates.

RQ2) Which regression model is the best at predicting average hotel room prices per night?
Random forest is the best performing model compared to ridge and lasso models. Random forest had the lowest RMSE (14.14), lowest MAE (8.91), and highest R^2 (0.818) which explains 81.8% of the variance in room prices on unseen test data, improvement over the regularised linear models at R^2 = 0.508.

RQ3) How does regularisation affect model performance and feature selection in predicting average hotel room prices per night?
Two regularised linear regression models were applied, L2 regularisation in ridge regression and L1 regularisation in lasso regression. Both ridge and lasso produced very similar performance metrics. The ridge model had an RMSE of 22.97, MAE of 17.57, and R^2 of 0.508 while the lasso model had an RMSE of 22.95, MAE of 17.54, and R^2 of 0.508. This shows that the linear form is constraint on the predictive accuracy of hotel room rates. Both models did shrink the coefficients to deal with possible multicollinearity issues but did not substantially improve predictive accuracy.

Lasso retained all 26 features and removed none. This indicates that all the features contributes meaningfully towards hotel room rates prediction and no feature was weak enough to be eliminated entirely at optimal regularisation strength. Lasso retained feature coefficient reveals the type of meal plan and type of hotel room to be largest coefficients while arrival date, lead time, and booking

history to be near-zero coefficients. However, ridge and lasso's R^2 of 0.508 compared to random forest's R^2 of 0.818 shows that regularisation alone is not sufficient to compensate for the limitation of linear assumptions when modelling a non-linear problem

3. Classification

3.1 Problem Statement

Hotel booking cancellations is one the most significant operational and financial challenges in the hospitality industry. Cancellations of reserved rooms are a loss of revenue for hotels. Over 40% of Booking.com revenue is cancelled annually (D-Edge, 2024). Thus, predicting whether a guest cancels their reservation is crucial. The study aims to identify the most accurate classification model for cancellation prediction, determine the booking features most discriminative for identifying cancellation-risk reservations, and compare the models.

3.2 Research Questions

RQ1) What is the best classification model to predict hotel room reservation cancellations?
RQ2) Which features are the most discriminative for identifying cancellation-risk reservations?
RQ3) How do different classification models compare in predicting hotel room reservation cancellation?

3.3 Dataset and Exploratory Data Analysis (EDA)

The "Hotel Reservation Dataset" from the previous regression task is utilised for classification as well. For classification, the target variable is booking_status.

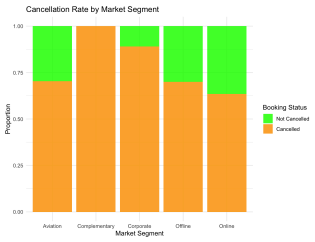


Figure 16. Cancellation rate by market segment

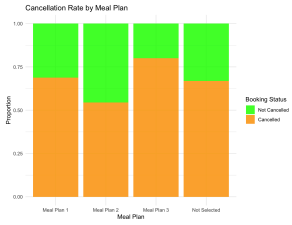


Figure 17. Cancellation rate by meal plan

Before applying classification models, EDA was conducted on the dataset. From Figure 16, the cancellation rates of five market segments are shown. The complementary segment had the highest cancellation at 100% followed closely by the corporate segment at approximately 90% cancellation rate. The high cancellation rate for the corporate segment could be due to last minute business plan changes causing hotel room reservations to be canceled frequently. The online segment had the lowest cancellation rate at 63%, however, this is still relatively high cancellation rates. Figure 17 shows the cancellation rate across four meal plans. Meal plan 3 has the highest cancellation rate at 80% while meal plan 2 has the lowest cancellation rate at 60%.

Before analysis, the dataset needs to be preprocessed. A 80:20 train-test split was performed to allow the training data to build the following models. 10-fold cross validation was used on the models to obtain a more accurate prediction of whether reservations are cancelled or not cancelled.

3.4 Analysis

3.4.1 Logistic Regression

Logistic regression is a supervised machine learning algorithm that predicts the probability of a binary outcome. It works by estimating the probability that an observation belongs to a particular class, such as when the probability is more than 0.5, the reservation is classified as cancelled, and when the probability is less than 0.5, the reservation is classified as not cancelled.

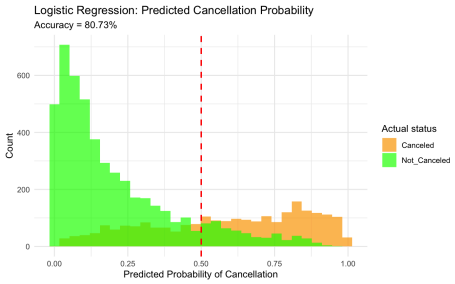


Figure 18. Distribution of Predicted Cancellation Probability

During the coding process, there were warnings of perfect separation for certain factor level combinations which means there were some factors that are able to perfectly predict booking status in the training data. The model was not affected by the warnings. From Figure 18, the model had an accuracy of 80.73% which indicates that this model is able to classify the majority of canceled and not canceled bookings well. The not canceled distribution is concentrated on the left side near zero, indicating that the model is relatively accurate in assigning low cancellation probabilities to most not cancelled bookings. The cancelled distribution is more spread out across the full range of predicted probability of cancellation where only slightly more canceled bookings are correctly assigned to predicted probabilities larger than 0.5 than incorrectly assigned.

3.4.2 K-Nearest Neighbours (KNN)

KNN is a classification algorithm that predicts the class of a new observation based on the classification of their nearest neighbours in the training data. KNN was evaluated using 5-fold cross-validation instead with three candidate k values (5, 9, 15) to balance computational efficiency with model selection reliability on a large dataset of 29,000 training observations. KNN measures similarity with Euclidean distance.

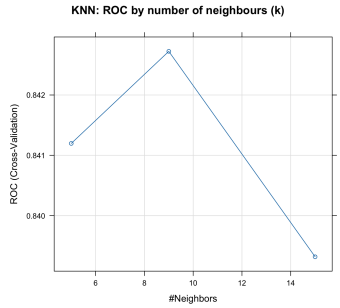


Figure 19. ROC by number of neighbours.

Figure 19 shows the cross validation ROC score for KNN across five k values. At k=9, the ROC score is the highest. From k=5 to k=9, the ROC score increased, suggesting that with more neighbours, the model was able to better predict cancellation or non-cancellation. However, from k=9 to k=15, the ROC score decreased, suggesting that the number of neighbours was too large and causing over-smoothing. Hence, the optimal k is 9.

3.4.3 Support Vector Machine (SVM)

SVM is a classification algorithm that finds the best possible boundary to separate two categories by maximising the gap between them. SVM searches for a decision boundary, known as a hyperplane, which separates canceled from not canceled bookings. The regularisation parameter C and kernel width sigma were tuned with 5-fold cross-validation due to the computational demands of SVM on a large dataset of 29,000 training observations.

3.4.4 Random Forest

Random forest was utilised similarly in classification as we did previously in regression.

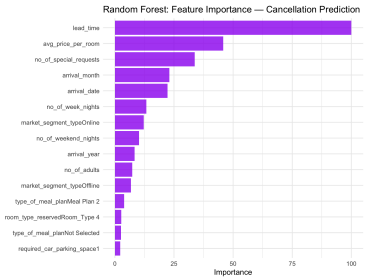


Figure 20. Feature Importance Ranking

From Figure 20, lead time is the most important predictor of booking cancellations at an importance score of 100. This is likely due to guests that have longer lead time are more likely to have the time to look for better alternatives, causing higher cancellations. The next most important predictor is the average price per room with an importance score of approximately 45. This could be due to guests that book more expensive rooms are more likely to cancel after finding cheaper alternatives to stay in. On the other hand, whether car parking space is required and the type of meal plan are the two least important predictors of booking cancellations. This indicates that guests that require or do not require car parking space behave similarly and thus, there is little predictive power. Similarly, the type of meal plan guests chose does not significantly affect whether they cancel their hotel booking.

3.5 Results

RQ1) What is the best classification model to predict hotel room reservation cancellations?

Model	Accuracy	Sensitivity	Specificity	F1	AUC
Logistic Regression	80.73	64.07	88.85	68.54	0.870
KNN	80.43	58.60	91.06	66.24	0.851
SVM	83.83	68.74	91.18	73.59	0.896
Random Forest	90.34	81.70	94.55	84.71	0.959

Table 3. Model Evaluation Performance

Random forest is the best performing model across all five evaluation metrics. It has the highest score in accuracy, sensitivity, specificity, F1, and AUC. With an AUC score of 0.959, it indicates very satisfactory discriminative ability where the model can reliably separate canceled and not canceled hotel room bookings.

RQ2) Which features are the most discriminative for identifying cancellation-risk reservations?
From Figure 20, lead time is the most discriminative feature with an importance score of 100, followed by average room price per night and the number of special requests at approximately 45 and 30 respectively. Monitoring lead times of guests would be helpful for managers to predict whether guests are likely to cancel or not cancel.

RQ3) How do different classification models compare in predicting hotel room reservation cancellation?

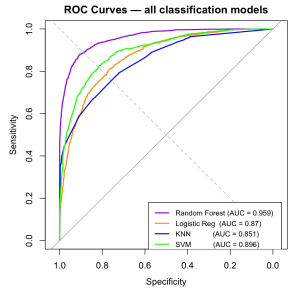


Figure 21. ROC Curves

Figure 21 shows clearly the best to worst performance among the four models. From the best to worst, the ranking is random forest, SVM, logistic regression, and lastly, KNN. The ROC curve for random forest is the closest to the top left corner which has a highest sensitivity of 81.40 and highest specificity of 94.55. SVM's ROC curve is the next best with the second highest score in accuracy, sensitivity, specificity, F1, and AUC. Logistic regression and KNN are the two lowest performers with overlapping at some portions of their curves. Overall, all four ROC curves sit above the diagonal line, indicating that all four models contribute predictive power for predicting cancellations of hotel reservations.

4. References

OECD (2026). *Inflation and cost of living*. [online] OECD. Available at: https://www.oecd.org/en/topics/inflation-and-cost-of-living.html.

Acropolium (2025). ▷ *Hotel Dynamic Pricing [A 2025 Guide for Hoteliers]*. [online] Available at: https://acropolium.com/blog/hotel-dynamic-pricing-solutions-development-trends-features-and-pitfalls/.

D-EDGE. (2024). *D-EDGE*. [online] Available at: https://www.d-edge.com/hotel-distribution-report-2024-have-direct-bookings-reached-a-peak/.

5. Dataset Links

Unsupervised Learning:
https://www.kaggle.com/datasets/mvieira101/global-cost-of-living/data

Regression and Classification:
https://www.kaggle.com/datasets/ahsan81/hotel-reservations-classification-dataset